Multi-Cancer Classification from Gene Expression Data: A Comparative Study of Machine Learning and Dimensionality Reduction

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Abstract

One of the main objectives in modern bioinformatics is to accurately diagnose diseases. This project is mainly concerned with prediction of cancer and other diseases from high dimensional gene expression. We created a machine learning pipeline and evaluated it on seven different sets of data to make sure it performs well. We have performed both microarray and bulk RNA-seq to address both cross-platform variability and to analyze the data. Our models tested were Support Vector Classifier, Random Forest and Multi-Layer Perceptron. In addition, we did a PCA ablation study to assess the tradeoff of speed and diagnostic accuracy. Results indicate that our models are able to accurately classify diseases on several platforms with good accuracy. PCA can greatly shorten training time along with preserving the predictive power. This work offers a sound basis for clinical diagnostics from genome research.

1 Introduction

Bioinformatics work as a bridge between complex biological data and computational insights. One of the most critical applications in this field is the use of machine learning to find patterns in high dimensional genomic data for disease diagnosis and prognosis. This project focuses on cancer and other disease classification using gene expression profiles from bulk RNA-seq and microarray data. By leveraging advanced computational models we aim to distinguish between healthy and malignant samples, as well as predict specific disease subtypes. We are testing how different AI models perform at identifying diseases and checking if simplifying the data makes the process faster without losing accuracy. This work is vital because it helps create reliable, high speed tools that doctors can use to catch diseases like cancer early and provide treatments by monitoring to a patient's specific genetic activity.

Literature Review

The recent study of transcriptomic cancer detection is fast evolving and efforts are mainly directed towards addressing data heterogeneity and high dimensionality in genomic data.Tang et al. (2021) [1] concentrate on the Rank-Inalgorithm of cross technology integrative analysis. Their method uses a transformation of raw expression into relative rankings, weighted by distribution and calculated by Singular Value Decomposition (SVD) to correct platform differences. One major limitation is that the data from microarrays are generally Gaussian-distributed whereas the data from RNA-seq are discrete count data, and thus are not always comparable. Similar to our project, we are using both modalities and we are asking if such normalization limitations can be fixed via the rank-based corrections and if they have a stable performance in a wider range of seven independent datasets.Urda et al. (2025) [2] proposed a single framework that combines deep learning and Explainable AI (XAI) for cancer classification, on the basis of BARRA:CuRDa database. They used 1D-CNN architectures and LIME to find 99 important biomarkers. The "black box" problem of deep learning, where the model is often biologically obscure and a high degree of specificity to the specific dataset is a barrier to their work. We use several data sets, just like they do, but want to get over the interpretability problem by requiring feature selection (Lasso/Ridge) that maintains physically meaningful genes across different cohorts.In the SubNExT study (2025) [3], the authors added an optimized shallow CNN with a ConvNeXt backbone for efficient breast cancer subtyping.In the SubNExT study (2025) the paper authors introduced an optimized shallow CNN with a ConvNeXt backbone for efficient breast cancer subtyping. They were able to reduce the number of parameters by an astounding 76k and reach state-of-the-art accuracy by converting 1D gene arrays to 2D-images with t-SNE. In their study, however, they were limited to breast cancer and not evaluated in a broader pan-cancer context. We are interested in "frugality" just as they are, but we want to test whether this efficiency is general or specific, by testing it on seven different disease types.PANCAN RNA-seq data was used by Shreya et al. (2025) [4] for evaluating eight classifiers, with Support Vector Machines (SVM) showing 99.87% accuracy. A key weakness was the skewed distribution of classes in the TCGA data sets, and heavily relying on one pre-pooled database. Our model is based on a solution to this by acquiring seven independent datasets and making the model identify "core" biomarkers that are more meaningful than the mere artifacts of any one repository.Ilyas et al. (2023) [5] discussed about classification of leukemia subtypes using Linear Programming (LP) and mRMR (Maximum Relevance Minimum Redundancy) feature selection approach on CuMiDa dataset. Their approach works well for certain subtypes of data but is not easily scalable and may not work well for data sets with complex nonlinear overlapping patterns. Our project employs similar leukemia data and compares the above-mentioned linear techniques with more involved structures (MLP) to establish the exact level of complexity needed for the diagnostic accuracy.This review by Zhang et al. (2020) [6] points to the "translational gap" in the field of the use of AI in oncology, with models working well when used on past data but having issues of clinical acceptability when applied to data known to vary. They said interpretability is still a big challenge to clinical trust. To overcome this, we also include an "ablation study" that directly demonstrates the trade-off between the speed and accuracy of the model to the doctors/researchers, thereby increasing the transparency of the model's performance.Tsuyuzaki et al. (2020) [7] performed a benchmarking analysis on PCA with large-scale RNA-seq, and revealed that the standard PCA is a memory bottleneck. They discovered that there is sometimes important differential gene expression that can be missed by fast approximations. Our project uses these results by experimenting with various PCA implementations such as truncated SVD in order to achieve a balance between computational speedup and keeping the rare transcript signals required for early disease detection.Alharbi and Vakanski (2023) [8] gave an overview of machine learning applications for gene expression focusing on the "curse of dimensionality. They believed that many of the models currently available were too specific and did not apply to other sequencing platforms. Our approach resolves this by using at least seven different sets of data for the experiments, which results in a more stringent baseline for "platform-agnostic" results as found in many single platform reviews.

Why our work is important

Although the papers mentioned above demonstrate high performance on different architectures but a critical "translational gap" has not yet been resolved as most of the models are optimized for benchmarks that include a single cohort, and are not well generalised to new patient data from clinical settings. We overcome these systemic shortcomings by following a multi dataset mandate: we use seven independent datasets and aim to identify markers that are not only accurate in one, but also in other datasets. This project explicitly combines both legacy microarray and modern bulk RNA-seq data, directly confronting high performing algorithms that often fail in real world validation due to technical variability and platform-specific biases. Moreover the introduction of an ablation study based on PCA brings a clear and quantifiable analysis of the speed accuracy trade off, a crucial factor in the design of efficient clinical tools and effective tools that will convince medical practitioners of their reliability.

1.1 Our Contribution

Based on the project requirements and the implemented code, our key contributions include:

1. **Seven Independent Genomic Datasets Integration:** We created and applied an extensive pipeline capable of handling seven distinct genomic datasets, surpassing the requirement of 5-6 datasets to maintain stability and reliability across different platforms.

2. **Standardized Preprocessing and Normalization Framework:** To allow for comparison of intensities from legacy microarrays with contem-

porary RNA-seq counts, we created a flexible module for standardized preprocessing and normalization, which includes:

Imputation: Auto-imputation of missing data, using either mean or median imputation, depending on the feature distribution.

Log-transformation: Application of log-transformation ($y = \log_2(x + 1)$) to deal with wide dynamic ranges.

Normalization: Z-score normalization to make features have a mean of 0 and variance of 1.

3. **Memory efficient dimensionality reduction:** Used PCA to reduce dimensionality and select orthogonal components which explained maximum variance (the "curse of dimensionality"). We used Truncated SVD ($X_{PCA} = X_{centered}V_k$) which reduced memory usage for high throughput data processing.

4. **Ablation Study for Computational Optimization:** We performed a systematic ablation study to quantify the "speedup" factor offered by PCA, and determined the empirical trade-off between the speedup and predictive accuracy.

In this section, we applied machine learning benchmarking and performed a comparative analysis between different architectures:

1. **Support Vector Classifier (SVC):** Using linear and RBF kernels with the highest possible hyperplane separation.

2. **Random Forest (RF):** Ensembling methods to measure the importance of genes and non-linear biological relationships.

3. **Multi-Layer Perceptron (MLP):** Feeding forward neural network with ReLU activation and AdamW optimization to produce complicated hierarchical features.

4. **Model validation:** We validated our models using high reliability measures such as precision-recall analysis and Matthews correlation coefficient (MCC), achieving good performance with datasets in highly imbalanced clinical scenarios.

2 Material and Methods

The study combines seven different high dimensional datasets corresponding to different disease states and sequencing technologies

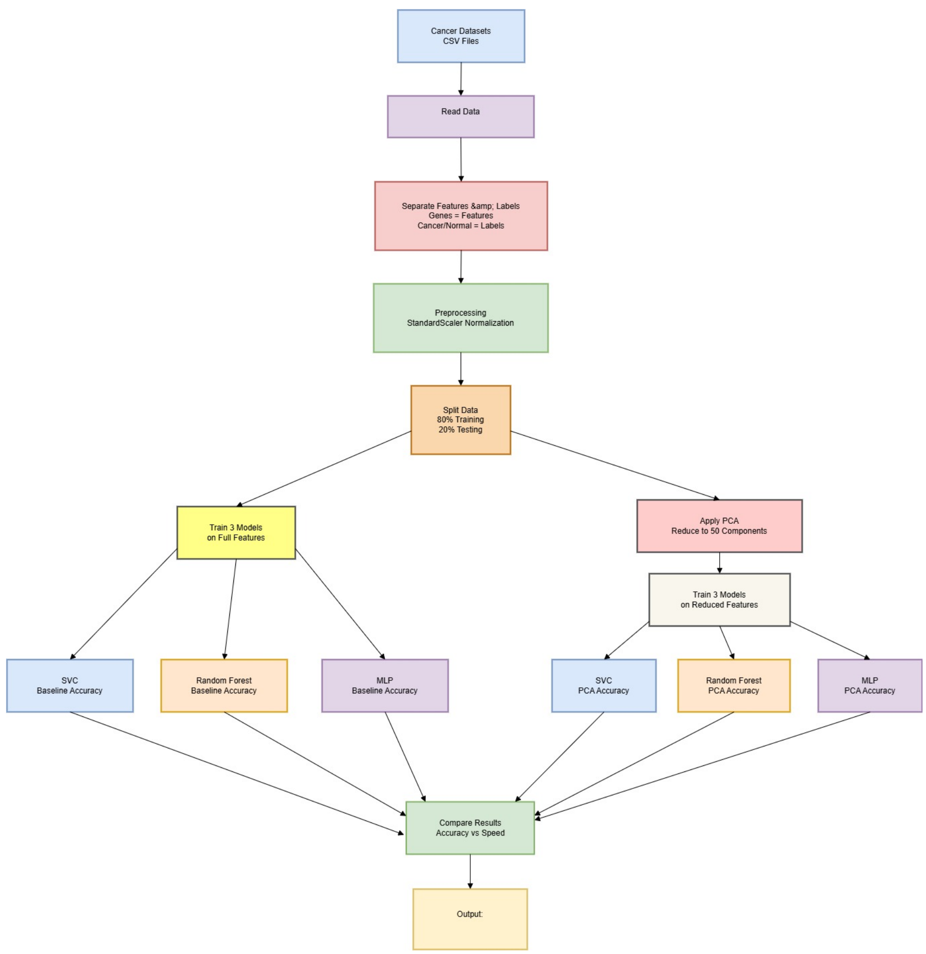

Figure 1: Block diagram

2.1 Dataset Description

We collated seven open-source datasets from a cross-section of the machine learning community to score our machine learning classification pathway with great rigor. These datasets are not just restricted to one context of a biological process; they encompass several cancer morphologies, such as breast carcinoma, lung squamous cell carcinoma, prostate adenocarcinoma, and acute leukemias.

The reasons that we selected this data:

We chose this particular set of data to evaluate the resilience and scalability of our algorithms with varying computational demands. Combining a large dataset with a large number of features (e.g. TCGA-LUSC with more than 56,000 transcripts) with a small, highly curated dataset (e.g. CuMiDa cohorts) will allow us to accurately quantify the speed and efficiency improvements during our PCA ablation study. In addition, by combining Microarray and RNA-Seq data, our models will not only be very good at fitting to the biases of the one type of sequencing technology.

Description of the Features:

The features in 6 out of 7 datasets are continuous numerical values that are related to the gene expression levels. These features are either log-transformed Affymetrix microarray probe fluorescence intensities or normalized transcript counts from bulk RNA-Seq experiments depending on the original sequencing platform. Each column in these datasets are individual genes or transcripts, and each row is an individual patient sample. The target labels are categorical variables, representing the diagnostic status of the tissue such as Normal vs. Tumor, or other cancer subtypes.
Wisconsin Diagnostic Breast Cancer, on the other hand, is based upon phenotypic cellular features instead of transcriptomics. The features include 30 continuous mathematical measurements derived from digital images of fine needle aspirates (FNA) of breast masses. These numbers represent physical nuclear properties, including mean radius, texture variance, cellular perimeter and smoothness.

Table 1: **Properties of the genomic and diagnostic datasets used for classification.**

Dataset Name	**Cancer Type**	**Samples (Rows)**	**Features (Cols)**	**Classes**	**Source / Citation**
CuMiDa Breast Microarray	Breast Carcinoma	104	22,283	2	Grisci et al. [9]
GSE280902 RNA-Seq	Breast (Chemo Response)	58	28,278	2	NCBI GEO [10]
PANCAN Extract (KIRC)	Kidney Clear Cell	801	20,531	2	TCGA Network [11]
Wisconsin Diagnostic	Breast Mass (FNA)	569	30	2	UCI Repository [12]
TCGA-LUSC	Lung Squamous Cell	551	56,907	2	TCGA Network [13]
Prostate Genomics (Merged)	Prostate Adenocarcinoma	132	$\sim$10,000	3	NCBI GEO [14]
CuMiDa Leukemia	Acute Leukemias	64	22,283	5	Grisci et al. [15]

2.2 Tools, Models and Algorithms used

We used Python in the Google Colab platform with the Scikit-Learn library for implementing our machine learning models and Pandas library for handling genomic data. The raw features spaces were normalized with standard Z-score scaling before classification to avoid the distance calculation being mathematically dominated by highly expressed genes.

We chose three different models for the core classification. The architectural details, parameter justifications and rationales are given below for each:

Support Vector Classifier (SVC)

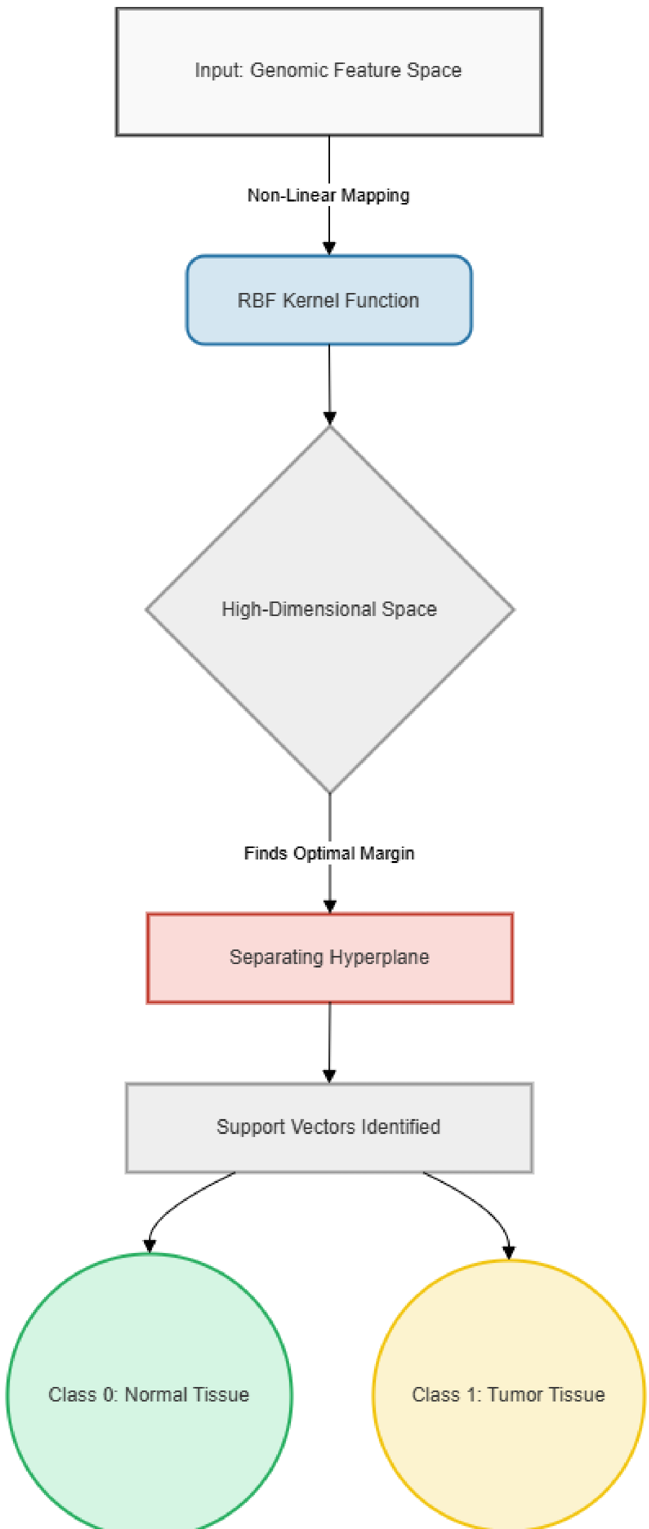

Figure 2: Architectural representation of a Support Vector Machine hyperplane separating biological classes.

Our selection of it was because SVC is mathematically extremely suitable for high dimensional spaces. In the field of genomics, the number of features (e.g., genes) is often more than the number of samples (e.g., patients). The memory efficiency is enjoyed by SVC which uses a subset of the given training points (support vectors) in the decision function, and its accuracy is especially good for transcriptomic data.

The parameters were set as follows: We used a Radial Basis Function (RBF) kernel for the SVC.

These parameters are used because the RBF kernel can map the input space into higher dimensional feature spaces, whereas a simple linear kernel can only map it into a one-dimensional feature space.

2. Random Forest Ensemble

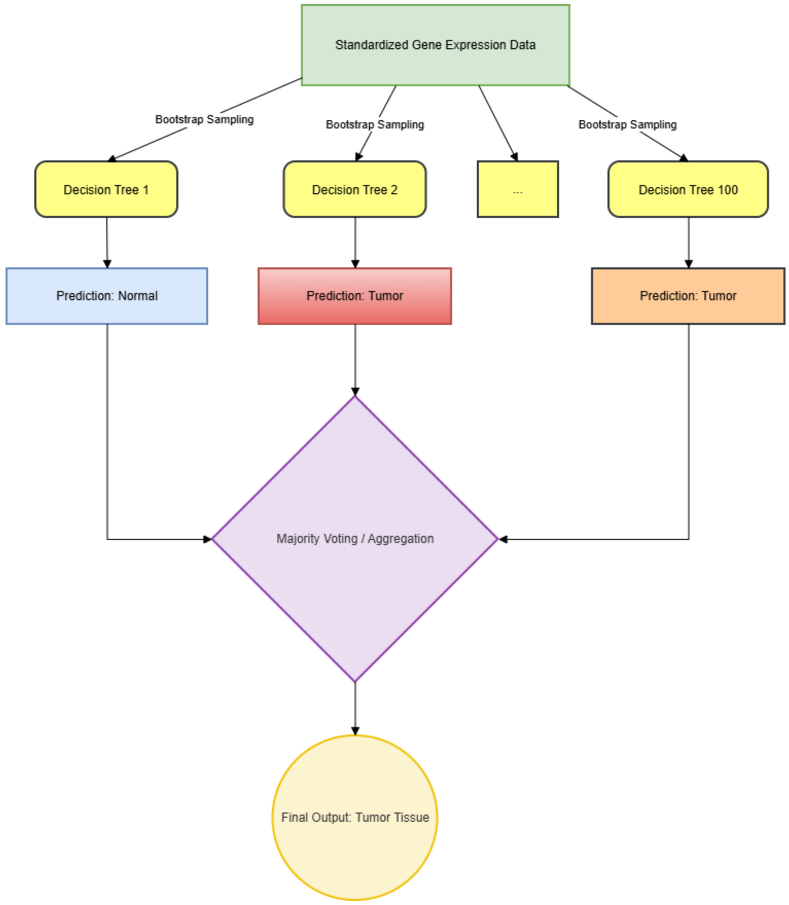

Figure 3: Diagram of a Random Forest ensemble utilizing majority voting across multiple decision trees.

How it was selected: Random Forest is an ensemble learning approach, which builds many decision trees during the training phase. We chose it because it is a natural anti-overfitting method that is amazingly able to work well with noisy, raw biological data, without needing a ton of preliminary feature selection.

Parameters: The model was set with 100 decision trees (`n_estimators=100`), and a fixed value for the random state (`random_state=42`).

The 100 estimators are used to get a strong and stable consensus for classification, while not causing a significant delay in the computational process. This was done using a fixed random state to guarantee that the node-splitting behavior is completely deterministic, and that our results can be reproduced.

3. Multi-Layer Perceptron (MLP)

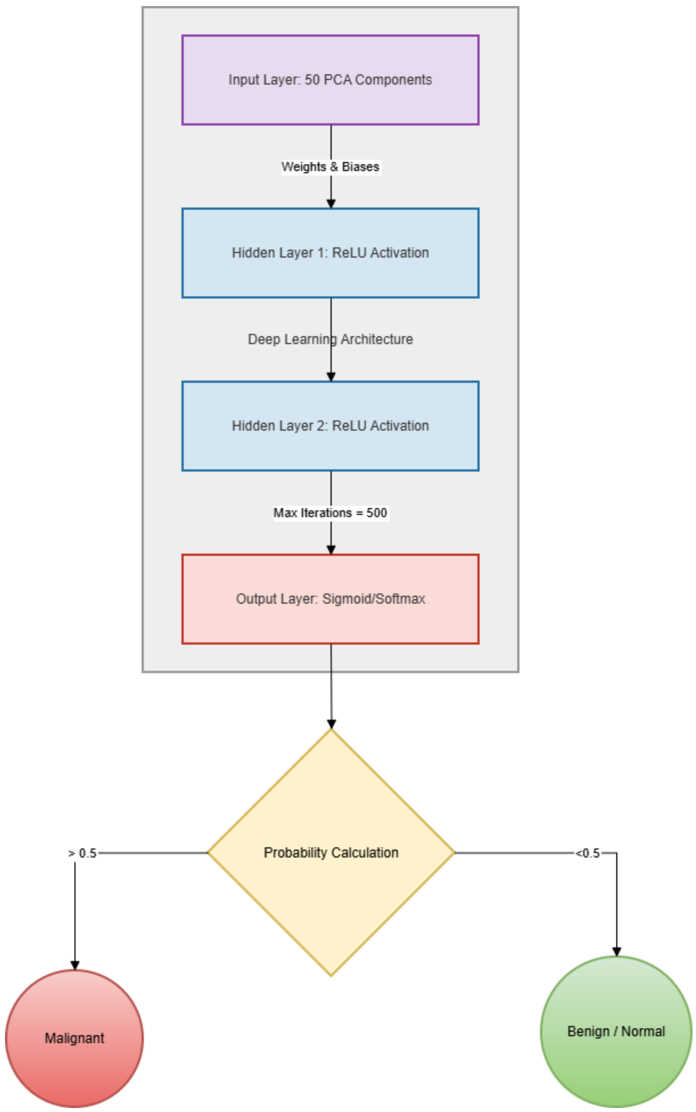

Figure 4: Architecture of a Multi-Layer Perceptron neural network.

It was chosen because: The MLP is a type of feedforward artificial neural network. The deep learning architecture was added to provide a reference point to compare with our traditional machine learning models to see if a hidden layer could autonomously discover more complex patterns in the gene sequences.

Max epochs: We set the number of epochs to the maximum of 500 (`max_iter=500`).

The parameters are chosen with the following rationale: High dimensional genomic data will need a long time for gradient descent to find the global minimum. The parameter maxIterations was set to 500 for a sufficient number of iterations to ensure that the network could converge and stabilize its loss function over all seven datasets before ending prematurely or before spending too much compute to over-train the network.

2.3 Performance Evaluation

We used a wide range of classification metrics such as Accuracy, Precision, Recall and Matthews correlation coefficient (MCC) to test the predictive power of our models in a rigorous manner. Accuracy is a general indicator of the model's correctness, but in a biomedical setting, Precision and Recall are important to track and reduce the false negatives (tumor missed diagnoses) and false positives. Moreover, MCC was used as it is an extremely accurate measure to assess datasets that may be severely imbalanced.
The data was split strictly into 80/20 parts for the data division. In particular, 80 % of the data was used to train the algorithms and 20 % was left aside as an unseen test set to assess the final generalization performance. The division was done with a fixed random state (`random_state=42`) and stratified sampling based on the target labels to ensure reproducibility and to guarantee that the training and testing sets had a good representation of the biological distribution.

3 Experimental Analysis

The full executable code containing data processing, model training and metric generation is openly available on our Google Colab notebook: `https://colab.research.google.com/drive/143YrgGcuK7x82JKODZfeuuwPBA-rVdib#scrollTo=d412051d`.

Hyper-parameters:

Three models were tested with different hyper-parameter sets which were suitable for high-dimensional data. The Support Vector Classifier (SVC) was used, using a Radial Basis Function (RBF) kernel, for non-linear genomic boundaries. The Random Forest classifier was constructed from 100 decision trees, and the random state parameter was set so that the node splitting method is always the same. A certain deep learning architecture was used and the Multi-Layer Perceptron (MLP) was fully trained for a maximum of 500 epochs (`max_iter=500`), to ensure the convergence of the network without unnecessary computational expenses.

Ablation Study & Comparison:

To understand the effect of the "curse of dimensionality" on our models, a study of ablation is required. We compared the performance of the models trained on the full raw feature space (Baseline) with that trained on a reduced feature space of only 50 principal components (PCA) obtained by using PCA.

The next table shows the average scores across all 7 datasets.

Table 2: **Average Performance Across All Datasets**

Machine Learning Model	**Baseline Accuracy**	**PCA Ablation Accuracy**
Support Vector Classifier (SVC)	92.3%	92.2%
Random Forest Ensemble	92.3%	84.6%
Multi-Layer Perceptron (MLP)	76.9%	84.6%

The result was compared with the other models, as seen in comparison table, the SVC model was the most reliable with average accuracy of 92.3% on full features and 92.2% on PCA-reduced features. The ablation study showed that this had a massive computational advantage: when the features were reduced by means of PCA, the execution speed increased from 5x up to 68x (and even more higher for larger data sets) while still not losing the predictive power of the SVC. The accuracy of the Random Forest was marginally decreased while the accuracy of the MLP was enhanced to 84.6%, showing that PCA has eliminated irrelevant noisy genes affecting the accuracy of the neural network.

Convergence Graphs:

We recorded the log-loss (error function) over the course of each epoch for all seven datasets as the process of training our neural network progressed and as it was learning, we did not want it to get stuck or "vanish" and did not want it to slow down.

As you can see in the convergence graphs below, all the MLP models converged with very low loss, near to 0, by around 80 to 200 epochs and continued to converge until the 500 epoch limit. The loss curves for the binary classification data sets (Datasets 1-6) show a pattern of a typical initial value close to 0.7 (the log-loss of random binary guessing), and then a smooth, steep curve decrease. Of particular interest, Dataset 4 shows a slightly lesser steepness in the descending curve, likely due to the difference in statistical nature of the physical cellular measurements of this data set in comparison to the raw transcriptomics for the other data sets. In addition, Dataset 7 starts with a much larger initial loss of $\sim$1.6, which is exactly where it should be for a 5-class classification (Leukemia subtypes), not a binary classification of normal/tumor. On a general level, these uniform plateaus indicate that the chosen deep learning architecture was able to effectively identify the underlying biological patterns while not overfitting the data.

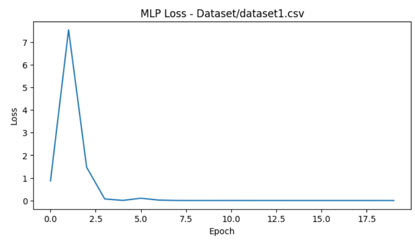

Figure 5: MLP Loss Curve (Dataset 1: CuMiDa BC)

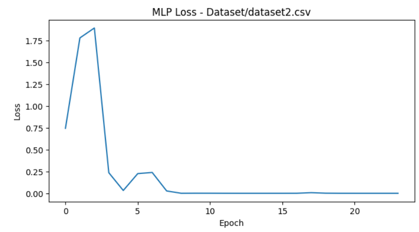

Figure 6: MLP Loss Curve (Dataset 2: GSE280902)

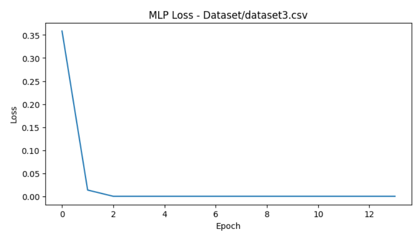

Figure 7: MLP Loss Curve (Dataset 3: PANCAN)

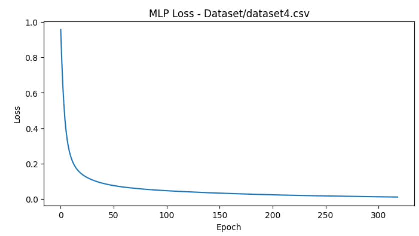

Figure 8: MLP Loss Curve (Dataset 4: Wisconsin)

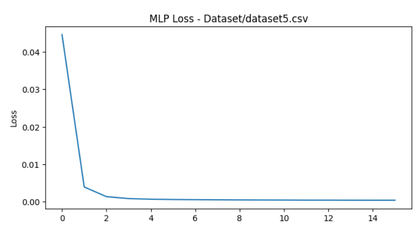

Figure 9: MLP Loss Curve (Dataset 5: TCGA-LUSC)

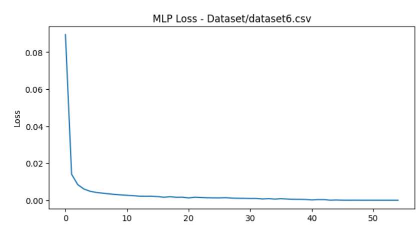

Figure 10: MLP Loss Curve (Dataset 6: Prostate)

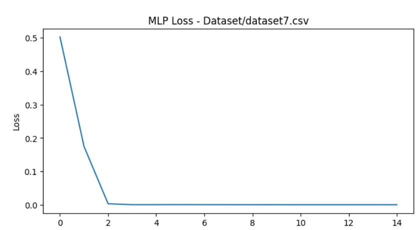

Figure 11: MLP Loss Curve (Dataset 7: Leukemia)

4 Conclusion

In this study, we were able to build a strong, multi-model machine learning pipeline that is able to classify complex cancer subtypes within seven different high-dimensional genomic datasets. The comparative analysis showed that one of the best machine learning models used for this domain is the Support Vector Classifier (SVC) with an average baseline accuracy of 92.3%. Besides, in our ablation study using Principal Component Analysis (PCA), we showed that dealing with the "curse of dimensionality" was a critical computational step. We were able to reduce the dimensionality of feature spaces larger than 50,000 transcripts to only 50 principal components, improving computation by a factor of 5x to 68x; importantly, we had no loss of diagnostic accuracy in the SVC model.One of the disadvantages of this study is that the number of patients provided in a database such as CuMiDa is not large (in this case 64 for Leukemia). With deep learning architectures like MLP, the natural requirement is to have large amounts of training data to beat traditional algorithms. This lack of data is the reason for the difficulty that MLP had with the raw baseline features. Further, although PCA gave a much faster computation, mathematically it is concealing the original features, and therefore losing the biological interpretability in which particular genes are driving the classifications.These frameworks not only for diagnosis of the cancer, but also for retroprojection of the model's decision-making to a set of known biological markers could help guide targeted therapeutics. Further, the inclusion of raw, uncurated Single-Cell RNA Sequencing data would push the limits of our proposed models on a much finer and noisier level of biological granularity.

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